

16. Evaluate :

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \cos^2 x \, dx$$

19. Find the integrating factor of the differential equation

$$x \frac{dy}{dx} = 2x^2 + y$$

24. Find :

$$\int \frac{\tan^3 x}{\cos^3 x} \, dx$$

27. Solve the following differential equation :

$$(1 + e^{y/x}) \, dy + e^{y/x} \left(1 - \frac{y}{x}\right) \, dx = 0 \quad (x \neq 0).$$

29. Evaluate :

$$\int_0^{\frac{\pi}{2}} \sin 2x \tan^{-1}(\sin x) \, dx$$

34. Using integration, find the area lying above x-axis and included between the circle $x^2 + y^2 = 8x$ and inside the parabola $y^2 = 4x$.

OR

Using the method of integration, find the area of the triangle ABC, coordinates of whose vertices are A(2, 0), B(4, 5) and C(6, 3).